Algebra & Equations

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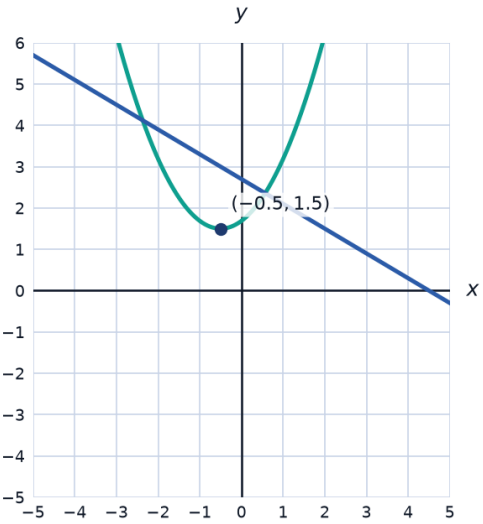
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Expanding and simplifying expressions**

Expand and simplify $3(2x - 1) - 2(x - 4)$.

2 **[Foundational] • Target: Factoring quadratic expressions**

Factor $x^2 - 49$ using the difference of squares.



3 **[Foundational] • Target: The quadratic formula and the discriminant**

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

4 **[Foundational] • Target: The quadratic formula and the discriminant**

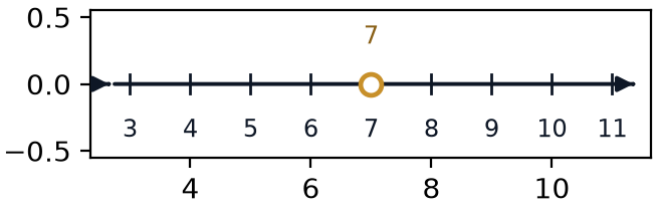
Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

5 **[Foundational] • Target: The quadratic formula and the discriminant**

Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

6 **[Foundational] • Target: Linear inequalities**

Solve $2x - 5 < 9$ and graph the solution set on a number line.



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7 [Foundational] • Target: Linear inequalities

Solve the compound inequality $5 < 2x + 1 \leq 11$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems of linear equations

Solve the system $x + y = 10$, $x - y = 2$.

9 [Intermediate] • Target: Systems of linear equations

This question has two parts. The sum of two numbers is 15 and their difference is 3.

a Write a system of two equations for the numbers x and y.

b Solve the system to find both numbers.

10 [Intermediate] • Target: Solving for a variable in a formula

The area of a triangle is $A = \frac{1}{2}bh$. Solve for h.

11 [Intermediate] • Target: Solving for a variable in a formula

The circumference of a circle is $C = 2\pi r$. Solve for r.

12 [Intermediate] • Target: Solving for a variable in a formula

The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t.

13 [Intermediate] • Target: Word problems: setting up and solving equations

This question has two parts. A rectangular garden has a length that is 4 meters more than its width w. Its area is 96 square meters.

a Write an equation relating w to the area.

b Solve for w, and state the garden's length.

14 [Intermediate] • Target: Word problems: setting up and solving equations

This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.

a Write a system of two equations for f and r.

b Solve the system to find the fixed fee and the rate per gigabyte.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: MCQ — Factoring quadratic expressions • Cross-Lesson: 9.01 Statistics & Probability

Find the value of m such that the quadratic equation $x^2 - 6x + m = 0$ has exactly one repeated real root.

16 [Challenging] • Target: MCQ — Solving quadratic equations by factoring • Cross-Lesson: 2.01 Functions & Transformations

If $(x - 2)$ is a factor of the polynomial $P(x) = 2x^3 - 3x^2 + kx - 10$, determine the value of k.

17 [Challenging] • Target: MCQ — The quadratic formula and the discriminant • Cross-Lesson: 3.01 Sequences & Series

For what value of k is the matrix $A = (k, 4; 3, 6)$ singular (has no inverse)?

18 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

Find the solution set of the inequality $|3x - 1| > 8$.

19 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

If $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$, determine the formula for $(f \circ g)(x)$ and state its domain.

20 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

Determine the horizontal and vertical asymptotes of the rational function $g(x) = \frac{4x - 1}{2x + 6}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — The quadratic formula and the discriminant

Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

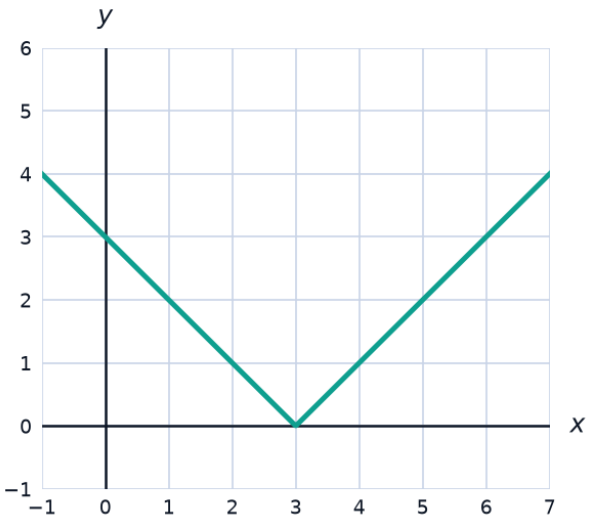
c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$

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22 [Foundational] • Target: MCQ — Absolute value equations

The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?



a. $x = 1$

b. $x = 3$

c. $x = -3$

d. $x = 0$

23 [Foundational] • Target: MCQ — Linear inequalities

Solve $3x - 7 < 14$, and identify the graph of its solution set.



a. $x < 7$

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

24 [Foundational] • Target: MCQ — Linear inequalities

Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$

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25 [Foundational] • Target: MCQ — Systems of linear equations

The sum of two numbers is 22 and their difference is 6. Find the two numbers.

a. 11 and 11

b. 14 and 6

c. 16 and 6

d. 14 and 8

26 [Foundational] • Target: MCQ — Solving for a variable in a formula

The perimeter of a rectangle is $P = 2l + 2w$. Solve for l.

a. $l = \frac{P}{2} - w$

b. $l = \frac{P - 2w}{2}$

c. $l = \frac{P + 2w}{2}$

d. $l = P - 2w$

27 [Foundational] • Target: MCQ — Solving for a variable in a formula

The formula for simple interest is $I = Prt$. Solve for r.

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

28 [Foundational] • Target: MCQ — Solving for a variable in a formula

The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C.

a. $C = \frac{5(F - 32)}{9}$

b. $C = \frac{5F - 32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F - 32)}{5}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Solving for a variable in a formula

The volume of a cylinder is $V = \pi r^2 h$. Solve for h.

a. $h = \frac{V}{\pi r^2}$

b. $h = V - \pi r^2$

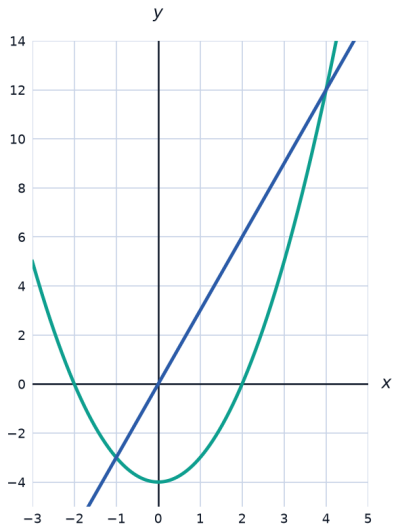
c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

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30 [Intermediate] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?



a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

31 [Intermediate] • Target: MCQ — Word problems: setting up and solving equations

A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

a. 12

b. 7

c. 9

d. 14

32 [Intermediate] • Target: MCQ — Word problems: setting up and solving equations

A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14

c. 54

d. 28

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33 [Intermediate] • Target: MCQ — Solving linear equations

[redacted] $ax^2 + bx + c = 0$ [redacted] $\Delta = b^2 - 4ac$ [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] $2x^2 - 8x + k = 0$ [redacted] k [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] For a quadratic equation $ax^2 + bx + c = 0$, if the discriminant $\Delta = b^2 - 4ac$ is equal to zero, it means the equation has one repeated real root. Given the equation $2x^2 - 8x + k = 0$, what is the value of k that results in this equation having one repeated real root?

a. 4

b. 8

c. 16

d. 32

34 [Intermediate] • Target: MCQ — Expanding and simplifying expressions

[redacted] $f(x) = ax^2 + bx + c$[redacted] [redacted] [redacted] [redacted] $f(0) = 5$ [redacted] $f(1) = 8$ [redacted] $f(2) = 13$[redacted] [redacted] [redacted] a. For the quadratic function $f(x) = ax^2 + bx + c$, if $f(0) = 5$, $f(1) = 8$, and $f(2) = 13$, find the value of a.

a. 1

b. 2

c. 3

d. 4

35 [Intermediate] • Target: MCQ — Factoring quadratic expressions

[redacted] [redacted] [redacted] [redacted] [redacted] [redacted] $x^2 - 5x + 6 = 0$[redacted] What is the sum of the roots of the quadratic equation $x^2 - 5x + 6 = 0$?

a. 6

b. -5

c. 5

d. -6

36 [Intermediate] • Target: MCQ — Solving quadratic equations by factoring

[redacted] [redacted] [redacted] [redacted] $x^2 + 5x + 3 = 0$[redacted] What is the discriminant of $x^2 + 5x + 3 = 0$?

a. 13

b. 19

c. 18

d. 11

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37 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

[redacted] If one root of a real quadratic equation is $2 + 3i$, what is the other root?

a. $2 - 3i$

b. $-2 + 3i$

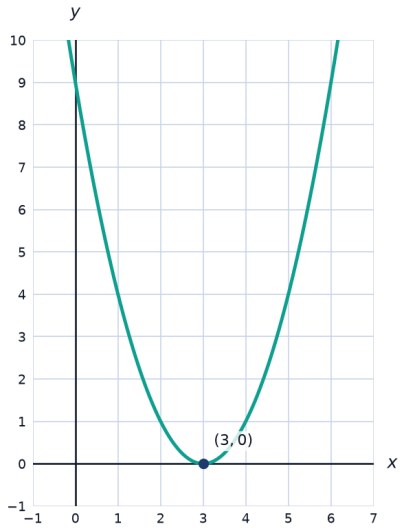
c. $-2 - 3i$

d. $3 + 2i$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

[redacted] The function $f(x) = x^2 - 6x + 9$ represents the trajectory of a projectile. What is the minimum height the object reaches?



a. 0

b. 3

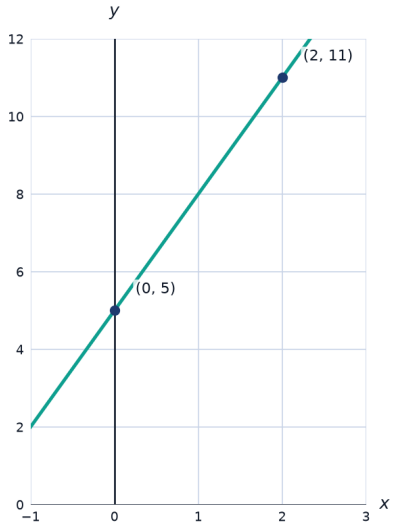
c. -3

d. 9

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39 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

[redacted] $y = mx + c$ [redacted] [redacted] [redacted] y [redacted] [redacted] $(0, 5)$ [redacted] [redacted] $(2, 11)$. [redacted] [redacted] m[redacted] [redacted] [redacted] [redacted] In the figure below, the straight line $y = mx + c$ represents a line that intersects the y-axis at the point $(0, 5)$ and passes through the point $(2, 11)$. What is the value of m, the slope of the line?



a. -3

b. 3

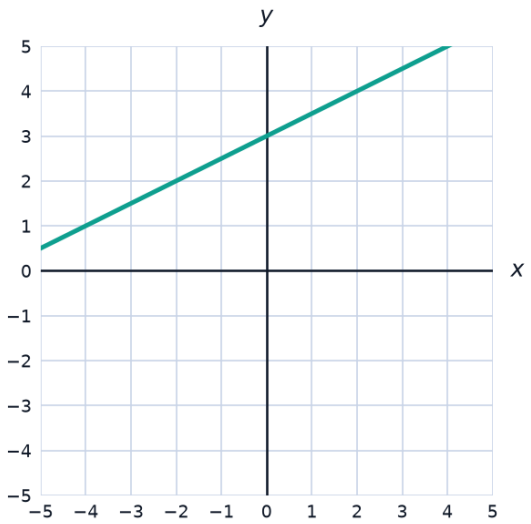
c. -2

d. 2

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40 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

[redacted] $y = mx + c$ [redacted] x [redacted] y. [redacted] [redacted] [redacted] $\frac{1}{2}$ [redacted] [redacted] $(4, 5)$[redacted] [redacted] [redacted] [redacted] [redacted] c. In the diagram below, the straight line $y = mx + c$ intersects the x and y axes. If its slope is $\frac{1}{2}$ and it passes through the point $(4, 5)$, find the value of the constant term c.



a. 3

b. -3

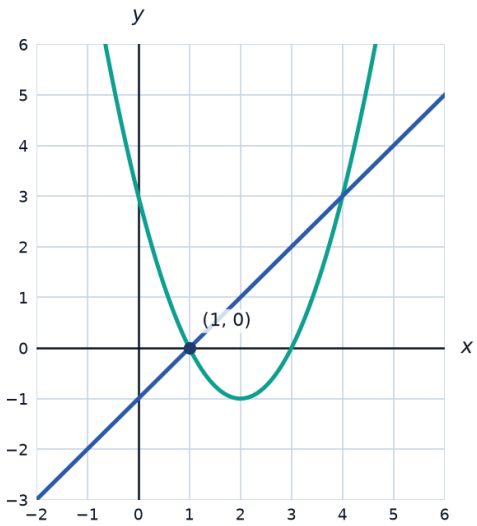
c. 1

d. -1

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41 [Challenging] • Target: MCQ — Rational equations • Cross-Lesson: 7.01 Integration & Applications

[redacted] $f(x) = x^2 - 4x + 3$ [redacted] $g(x) = x - 1$. [redacted] [redacted] (x, y) [redacted] $y = 0$[redacted] The graph shows the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. What are the coordinates of the intersection point (x, y) where $y = 0$?



a. $(1, 0)$

b. $(3, 0)$

c. $(0, 3)$

d. $(0, 1)$

42 [Challenging] • Target: MCQ — Solving for a variable in a formula • Cross-Lesson: 8.01 Exponentials & Logarithms

[redacted] x [redacted] y [redacted]: $\{x^2 + y^2 = 13\ xy = 6$ [redacted] $x + y$[redacted] [redacted] x [redacted] y [redacted] [redacted] If x and y are real numbers satisfying the following system of equations: $\{x^2 + y^2 = 13\ xy = 6$ What is the maximum possible value of the expression $x + y$, given that x and y have the same sign?

a. 3

b. 5

c. 6

d. 7

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43 [Challenging] • Target: MCQ — Linear-quadratic systems • Cross-Lesson: 9.01 Statistics & Probability

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] L_1 [illegible] L_2 [illegible] [illegible] T. [illegible] [illegible] [illegible] [illegible] α [illegible] 60° [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] β[illegible] [illegible] [illegible] β [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] α[illegible] In the figure shown, you have two parallel lines L_1 and L_2 intersected by a transversal T. If the measure of angle α is 60°, what is the measure of angle β, given that β is the consecutive interior angle to α?

a. 30°

b. 60°

c. 120°

d. 90°

44 [Challenging] • Target: MCQ — Word problems: setting up and solving equations • Cross-Lesson: 2.01 Functions & Transformations

[illegible] [illegible] $5x - 7 = 3x + 9$[illegible] [illegible] [illegible] [illegible] x[illegible] [illegible] [illegible] x [illegible] [illegible] [illegible] If $5x - 7 = 3x + 9$, what is the value of x, given that x is a real number?

a. 7

b. 8

c. 6

d. 9

45 [Challenging] • Target: Solving linear equations • Cross-Lesson: 3.01 Sequences & Series

[illegible] [illegible] [illegible] $3x - 5 = 7$[illegible] [illegible] [illegible] x[illegible] To solve the equation $3x - 5 = 7$, what is the value of x?

a. x = 4

b. x = 3

c. x = 2

d. x = 5

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